IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DISCRETE-EVENT SYSTEMS

Wednesday, 6 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : D. Angeli
Second Marker(s) : E. Kerrigan

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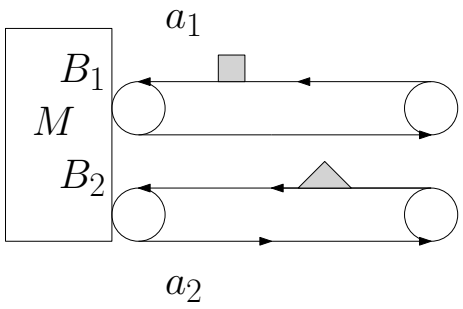


Figure 1.1 Machine M, with arrivals a_1, a_2 and associated buffers B_1, B_2

1. A machine M operates by assembling two different types of pieces (type 1 and 2, respectively) into a single object. Arrivals of pieces of type $i = 1,2$ are detected by a corresponding sensor and modeled as a_i events. Each type of piece can be stored in a dedicated buffer (B_1 and B_2, respectively) of infinite capacity. Both buffers start with the same number of pieces (of type 1 and 2), and, for the sake of simplicity, we assume that the machine never runs out of them, (i.e. buffers are never empty regardless of the production rates).

   a) A supervisor inhibits arrivals of pieces of type 1, whenever the buffer B_1 has one piece more than buffer B_2. Similarly it inhibits arrivals of type 2 whenever buffer B_2 has one piece more than buffer B_1. Model the relative occupancy of the buffers B_1 and B_2 by means of a finite deterministic automaton G which keeps track of the difference of pieces of type 1 and 2 stored at the machine buffers. [4]

   b) Occasionally, when buffer B_1 has one piece more than buffer B_2 the supervisor fails to inhibit a_1 arrivals and a piece of type 1 physically enters the buffer B_1 without being detected by the sensor for a_1 events. Such event is modeled as an unobservable fault f. After the fault, operations continue as normal (i.e. arrivals of type a_1 are inhibited and arrivals of type a_2 are still possible). Extend the previous FDA by keeping track of the relative number of pieces of type 1 and 2 in the presence of f events, and denote the corresponding model as G_F. [4]

   c) Design a labelling automaton G_L which keeps track of the first occurrence of f events, to be later used for the design of a diagnoser automaton. [2]

   d) Compute the parallel composition $\tilde{G} = G_F || G_L$. [3]

   e) Replace, in $\tilde{G}$, f events by ε events (empty events). Design the Observer Automaton $\text{Obs}(\tilde{G})$ for the resulting non-deterministic automaton. [5]

   f) What is the state reached after the word $w = a_2a_1a_1a_2a_2a_2a_1a_1$ is detected by the observer automaton? What does it mean for the original system (has fault event occurred? What are the possible difference values between the number of pieces in the buffers B_1 and B_2?) [2]

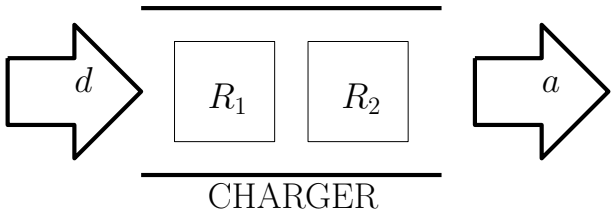


Figure 2.1 Robots and FIFO charging area

2. A factory operates with two mobile robots, R_1, R_2. Each one of them operates in one of two possible states: O, for operational, when the robot is able to move around the factory, and C for charging, during which the robot is constrained to be within the charging area. When robot R_i runs out of energy a 'docking' event d_i is triggered and the robot enters the charging area (its state then becomes C). When robot R_i has accumulated enough charge while at the charger an activation event a_i occurs, and the robot is able to leave the charger area.

   a) Model each robot, R_i by a Finite Deterministic Automaton G_i, $i = 1,2$. Sketch the transition diagram assuming that both robots are in initial state O. [3]

   b) Compute the concurrent composition $G = G_1||G_2$. [3]

   c) Assume next that a strict First In First Out policy needs to be enforced at the charging unit, (see Fig. 2.1), as robots are supposed to enter the charging area from one side and go out on the opposite side. Model this specification as the generated language of a suitable Finite Deterministic Automaton G_K. [4]

   d) Assume next that events d_1 and d_2 are uncontrollable. Is the considered specification controllable? (Justify your answer). [3]

   e) Assume that event a_1 is uncontrollable, i.e. $E_{uc} = \{a_1\}$. Is the specification $\mathscr{L}(G_K)$ controllable ? If not, find an admissible supervisor S such that $\mathscr{L}(S/G)$ implements the maximal controllable sublanguage $\mathscr{L}(G_K)^{\uparrow C}$. [5]

   f) In a situation in which 3 autonomous Robots are operating within the charging station according to a FIFO policy, how many states do you foresee being needed to represent such policy as a state-splitting specification, for the state CCC (in which all robots are charging)? [2]

3. Consider the following common biochemical interaction network. An enzyme E is free to move within the cytoplasm of a cell where it may bind to a substrate S (when both free enzyme molecules and substrates are available), and form, as a product of this reaction, a complex Enzyme-Substrate (ES). Occasionally the complex ES splits up again as $E + S$ and the enzyme molecule unbinds, freeing up the substrate S at the same time. The complex ES is otherwise likely to combine with a phosphate group Ph and (at the same time) split up again as the enzyme E plus a product P (the latter is basically a version of S with an added phosphate group).

a) Model this network of biochemical interactions by means of a Petri Net N, whose marking should track the number of Enzyme molecules E, number of available Substrates S, number of ES complexes, number of products P and number of phosphate groups Ph. Make sure that all reactions mentioned above appear as suitable transitions joining the corresponding input (reactants) and output places (products). [6]

b) Find the incidence matrix C of the considered Petri Net. [2]

c) Find, if any, the P-semiflows of the network and provide a physical interpretation of their meaning. [4]

d) Assume next that an additional reaction is considered, namely a degradation, by which P transforms into S. Denote this modified net as $\tilde{N}$. Explain why and how one of the P-semiflows previously identified is no longer a P-semiflow for the modified Petri Net. Is this vector still a P-invariant vector of some kind ? (justify your answer). [3]

e) Discuss the structural conservativeness of the original net N and of the modified network $\tilde{N}$. [3]

f) Discuss the structural boundedness of the original net N and of the modified network $\tilde{N}$. [2]

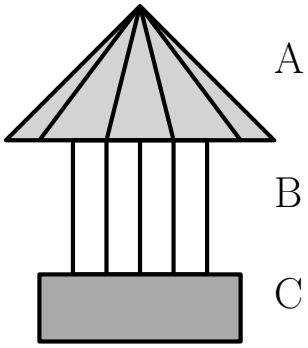


Figure 4.1 Toy house parts

4. A machine M operates by assembling parts A (roof), B (walls) and C (pedestal) to form a toy house, see Fig. 4.1. Each of the gluing and arrival processes which are entailed in the production have an exponentially distributed life-time. The machine can operate according to 2 different gluing paths (listed below), each one terminating with a complete toy house:

1. M receives parts A and B already glued together, then it completes the house by gluing C, (at rates λ_1 and λ_2 respectively);

2. M receives parts B and C already glued together, then it completes the house by gluing A, (at rates λ_3 and λ_4 respectively);

Assume that the machine M is initially idle, and waits for arrivals of glued pieces AB, or BC, respectively (each at the corresponding arrival rate). Then, as soon as these arrive, further arrivals are inhibited and the gluing process with pieces of type C and A (respectively) may begin (assuming that individual parts A and C are always available at the machine and that the machine M chooses the path 1 or 2 depending on which arrival comes first). The gluing processes happen at the rate specified above.

a) Model the operation of the machine M as a Markov chain. With initial idle state (waiting for the parts to arrive) and final absorbing state corresponding to the completion of the toy house. [5]

b) Write the Kolmogorov equations which characterize the evolution in time of the probability vector π. [3]

c) Compute the average time it takes from when the machine starts out in the idle state, to when the toy house is completed. [5]

d) Modify the Markov chain in such a way that the transitions leading to the terminal state (toy house completed) are redirected towards the initial state (idle machine), so as to model a machine M continuously operating on such a cycle. Write the equations of the modified Markov chain. [2]

e) Compute the fraction of time spent with the machine in the idle state. [3]

f) Explain whether the previous quantity is the expected fraction of time spent in the idle state, or the fraction of time spent in the idle state with probability 1. [2]